Metadata for East Kilbride (EAS), South Lanarkshire, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field at the remote site (EASR) and the natural magnetic field plus the field created by GIC at the underline site (EASU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details.
Location (in lat/long): underline station	55.706125 N, -4.149695 E
Location (in lat/long): remote station	55.706651 N, -4.152130 E
Distance between DMM sites	200 m
Description of site (i.e. open moorland)	Unused open fields
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	14/08/2019
Removal date	02/06/2021
Number of days collected	EASR: 491 EASU: 557
Data sampling rate	1 Hz
Notes of known data gaps	EASR: Data gaps: 01/01/-06/01/2020, 09/09/2020-05/01/2021 EASU: Data gaps: 01/01/-03/01/2021
Qualitative estimate of site quality	Very good
Orientation angle of HV wires relative to true north	+24.8° (i.e. lines run approximately NNE-SSW)
Alignment of magnetometers	Magnetic north as of 14/08/2019
Voltage level of the lines	400 kV (Strathaven – Kilmarnock)
Height above the ground of each wire (m)	Lower: 14.71 m, Middle: 22.22 m, Upper: 30.05 m
Underline magnetometer	Central between both circuits, 9 m to each circuit
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421